

# Dana Alrijjal

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## Education

**Effat University**, Bachelor of Science in Computer Science

Aug 2022 – 2026

- GPA: 3.97/4

**Coral International School**, American High School Diploma

Sept 2008 – May 2022

- GPA: 4/4

## Certifications

**KAUST Academy – AI Specialization Stages 1–3 Certified**

Jan 2026

- Completed advanced coursework in Computer Vision, NLP, Deep Learning, Reinforcement Learning, and Graph Neural Networks through a competitive multi-stage program.

## Experience

**Associate**, Cedge

June 2026 – Present

- Contribute to the development and improvement of the company's internal systems and digital solutions.
- Conduct business research and develop proposals to support strategic initiatives and technology-driven solutions.

**Special Educational Need Tutor**, Reach Inclusion

Aug 2022 – Oct 2023

- Adapted teaching methods to meet individual learning needs, ensuring an inclusive and supportive educational environment.

## Papers

**Alrijjal, D., & AlDaghma, J. (2024). Advancements in Kernel Concurrency: Leveraging Machine Learning for OS Innovation.**

- Reviews machine learning approaches for kernel concurrency, including bug detection, test prioritization, and adaptive synchronization.

**Alrijjal, D., & AlDaghma, J. (2025). Evaluating Digital Twin Technology for Proactive Cybersecurity Defense.**

- Analyzes Digital Twin applications in cybersecurity, demonstrating 97.5% detection accuracy and 1.5s latency in simulated environments.

**Alrijjal, D., & AlDaghma, J. (2025). A Multi-Layered Adaptive Framework for Adversarially Robust AI in Cybersecurity Applications.**

- Proposes an adaptive defense framework combining adversarial detection, retraining, and explainable auditing for cybersecurity AI.

## Projects

**AegisAI – LLM-Driven Penetration Testing Framework**

- Developed an AI penetration-testing framework using LLM-driven attack generation, Digital Twin sandboxing, automated reconnaissance, and adversarial evaluation across AI/ML, LLM, RAG, and multimodal systems.

**Adversarial-Resilient Big Data Pipeline for Cyber Threat Intelligence**

- Built a 7-stage Apache Spark pipeline processing 11K+ CTI records, combining anomaly detection, adaptive data repair, ML classification, and audit while reducing training noise by 51.7%.

**Adversarial-Robust NLP for Cyber Threat Intelligence (CTI)**

- Fine-tuned DistilBERT for CTI classification with FGSM adversarial training and evaluated robustness using Clean Accuracy, Macro F1, Robust Accuracy, and Attack Success Rate (ASR).

## Honors and Awards

**Dean's List, Effat University**

2022–2026

- Honored for exceptional academic achievement and consistent excellence in coursework.

**Academic Merit Scholarship, Effat University**

2022–2026

- Awarded 50% scholarship based on academic performance.

## Technologies and Skills

**AI / ML:** Python, PyTorch, TensorFlow, HuggingFace, NLP, Transformers, Adversarial ML

**Data:** Apache Spark, SQL, Pandas, NumPy, Matplotlib

**Cybersecurity:** Cyber Threat Intelligence, Digital Twins, Adversarial AI

**Programming / Web:** C++, Java, HTML, CSS, JavaScript, PHP

**Tools:** Git, Jupyter Notebook, Scikit-learn, Cisco Packet Tracer